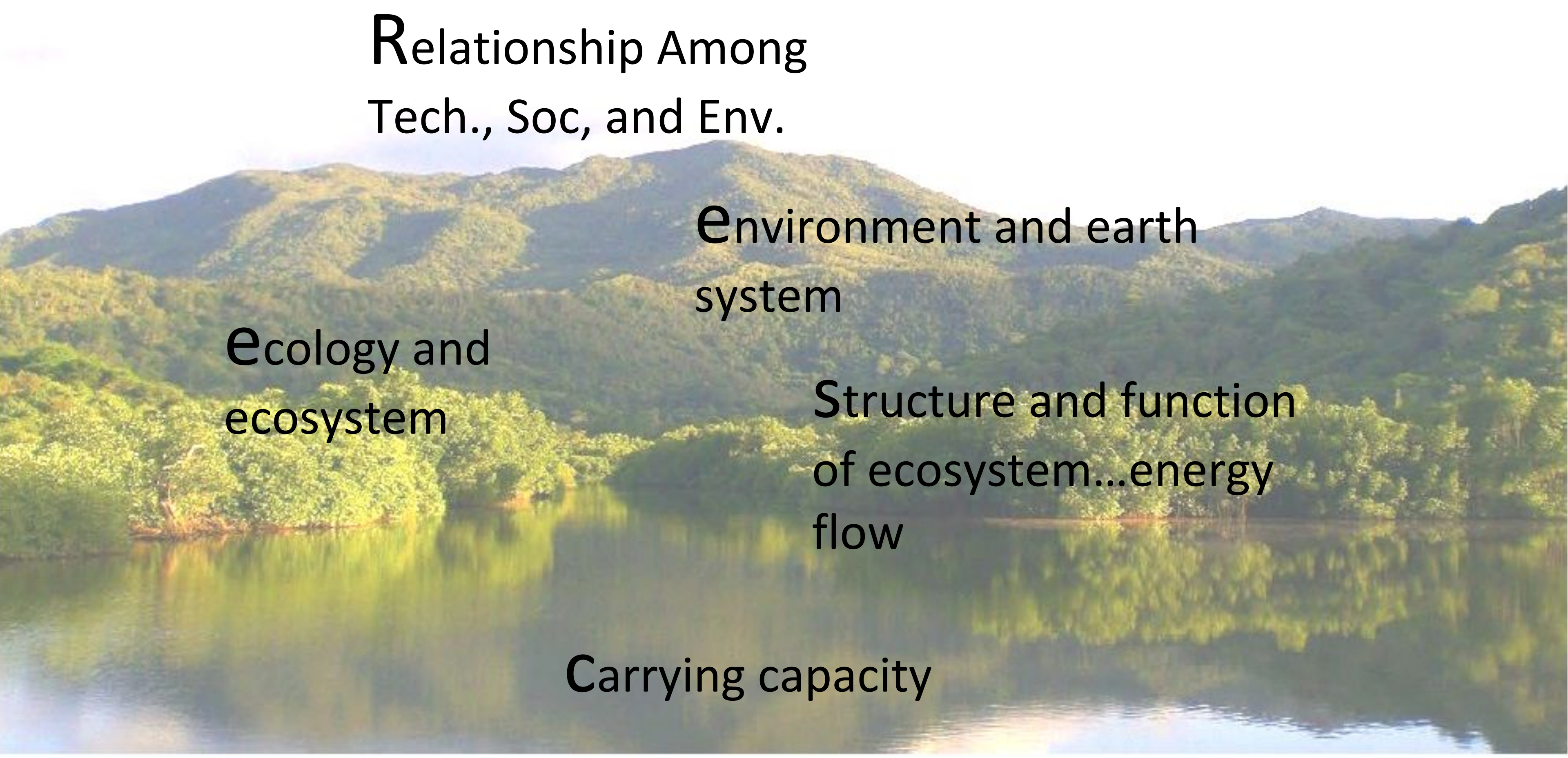
Course Title: Technology, Environment and Society

HUM 4147

What we went through…



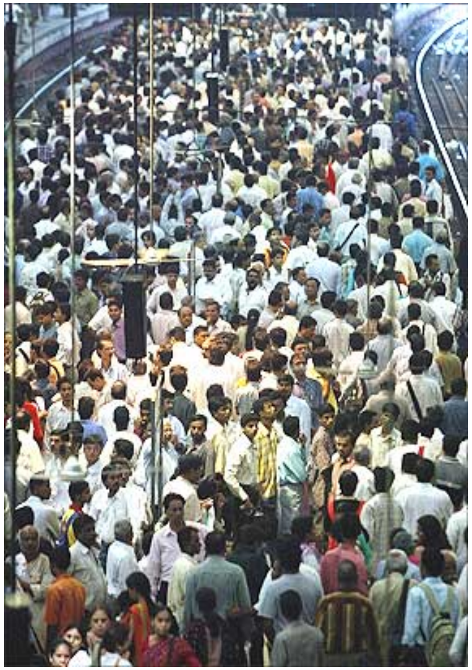
Global Environmental Picture

Major Global Environmental Concerns:

- Population Growth
- Soil Degradation
- Atmospheric Change
- Biodiversity Loss

Global Environmental Concerns

Population Growth



Biodiversity Loss

Atmospheric Change

Soil Degradation

Global Environmental Concerns

Population Growth

Population

total human inhabitants of a specified area, such as a city, country, or continent, at a given time. Population study as a discipline is known as demography.

Deals with

It is concerned with the size, composition, and distribution of populations; their patterns of change over time through births, deaths, and migration; and the determinants and consequences of such changes.

Why do you need to know?

knowledge on Population studies is important for planning, particularly by governments, in fields such as health, education, housing, social security, employment, and environmental preservation. Such studies also provide information needed to formulate government population policies, which seek to modify demographic trends in order to achieve economic and social objectives.

Global Environmental Concerns

Population Growth

Robert McNamara(1973), wrote *One Hundred Countries, Two Billion People*

rapid population growth was a threat that would have
"catastrophic consequences"

James Wolfensohn,
World Bank president from 1995 to 2005

limited reference to
population growth

Robert Zoellick
delivered a speech identifying
six strategic themes to meet global challenges. Reducing population was not one of them; in fact, "population growth" is not mentioned even once

Robert McNamara, *One Hundred Countries, Two Billion People* (New York: Praeger, 1973), p. 31.

Global Environmental Concerns

Population Growth

last Ice Age, about 13,000 years ago, when humans on all continents were still living as hunter–gatherers, or 12,000 years ago with the first signs of agricultural settlements, or 7,000 years ago with the first indications of urbanization.

From the beginning of human settlements, it took more than
10,000 years for the world's population to reach 1 billion in 1820.

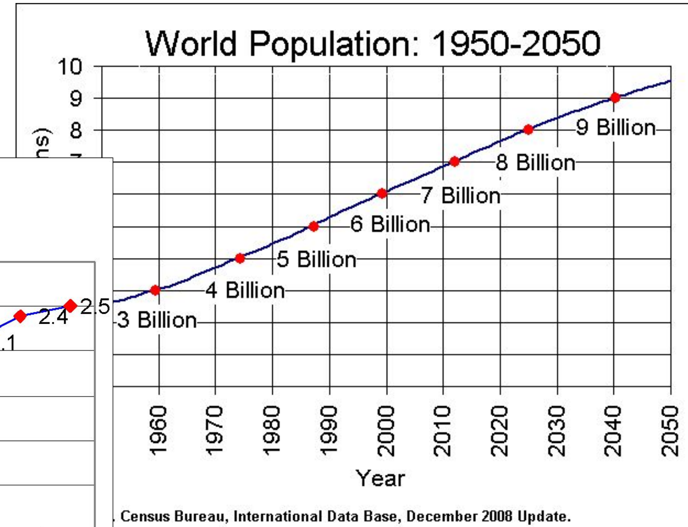
next billion were added in only about 110 years and, for the last four decades, the world added
1 billion people every 12 to 15 years. No wonder McNamara was so worried.

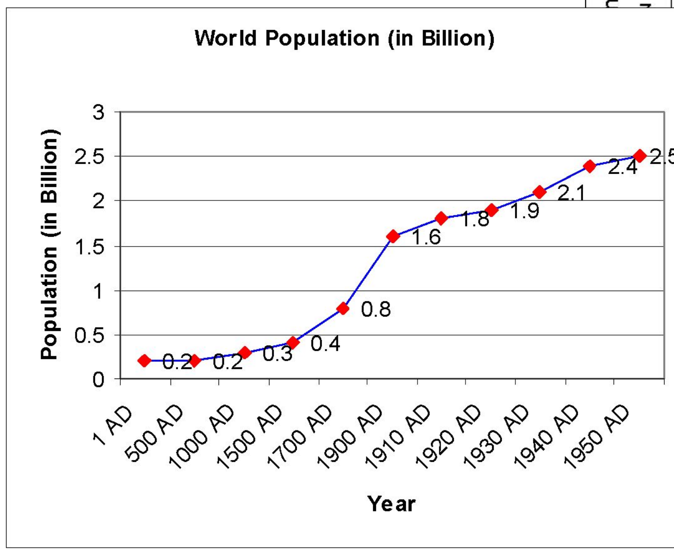
During the Industrial Revolution (late eighteenth and early nineteenth centuries), the food supply grew and became more reliable. The death rate fell, life expectancy increased, and population growth gradually accelerated.

The Industrial Revolution, which marked the start of modern economic growth, further expanded the earth's **population-carrying capacity**.

Global Environmental Concerns

Population Growth

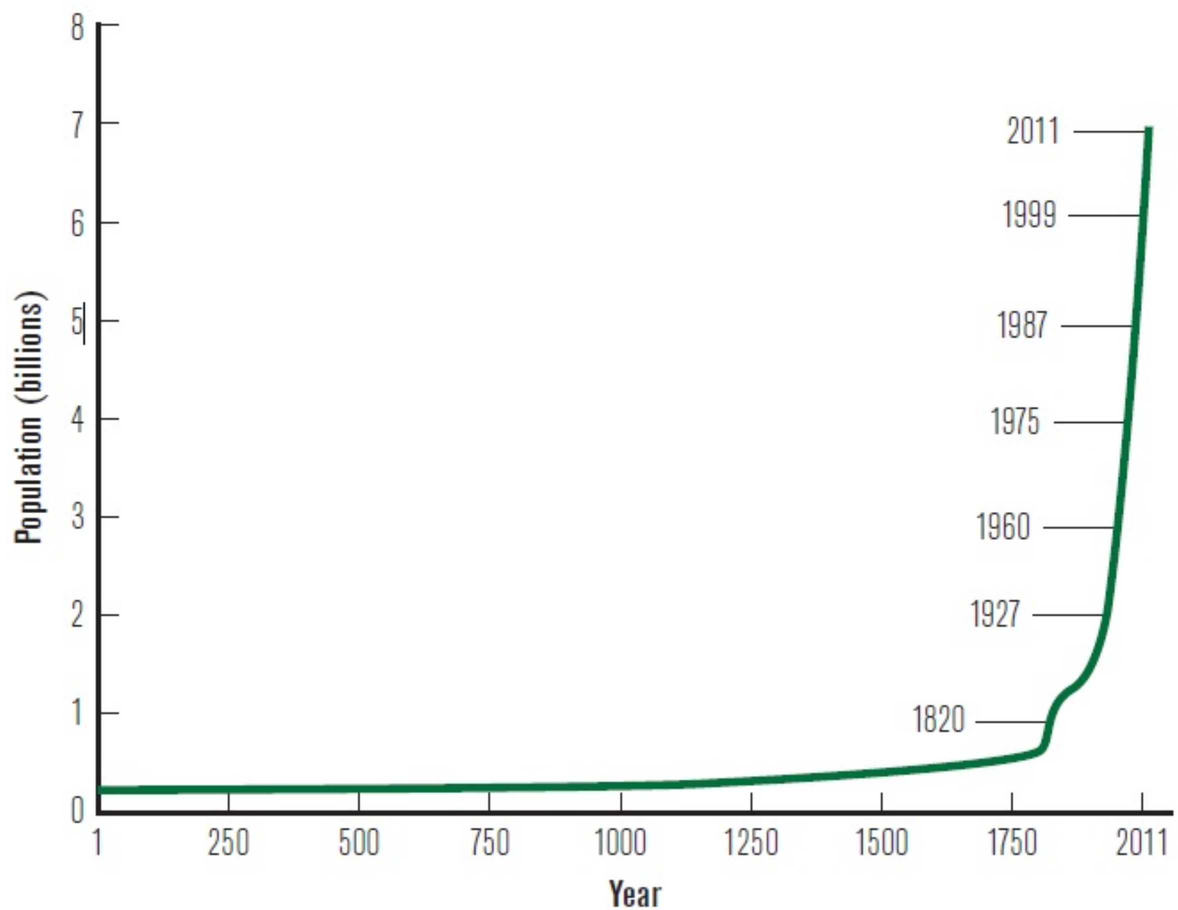




- four or five children are born in every second
- two other people die in same time
- nearly 2.5 more humans per second added to the world population.
- This means, we are growing almost 83 million more people per year.

Global Environmental Concerns

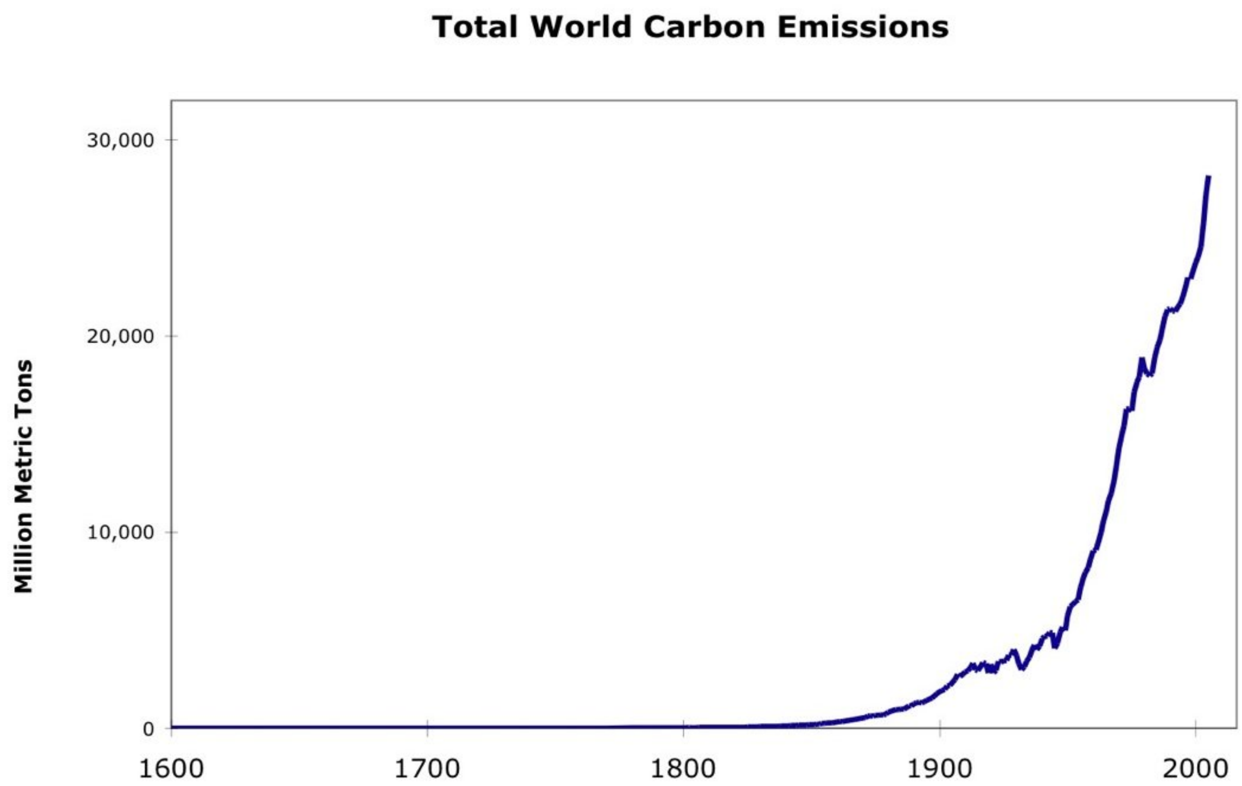
Population Growth



Don’t know what made this chart exploded since 1900. Charts are so exponential and one of the major cause of Global Warming.

Global Environmental Concerns

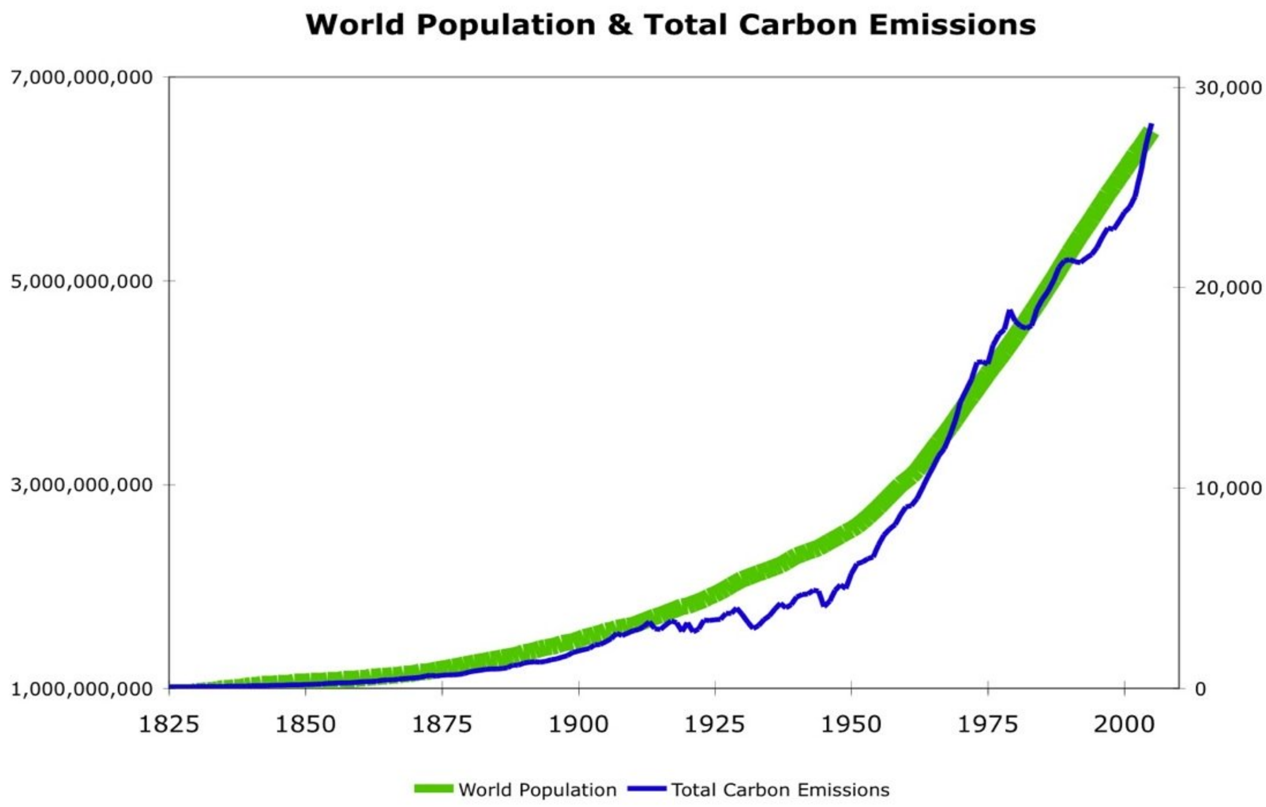
Population Growth



A significant Hockey Stick. Worldwide Total Carbon Emissions have experienced exponential growth. The chart shows historic Total Annual Worldwide Carbon Emissions since 1600.

Global Environmental Concerns

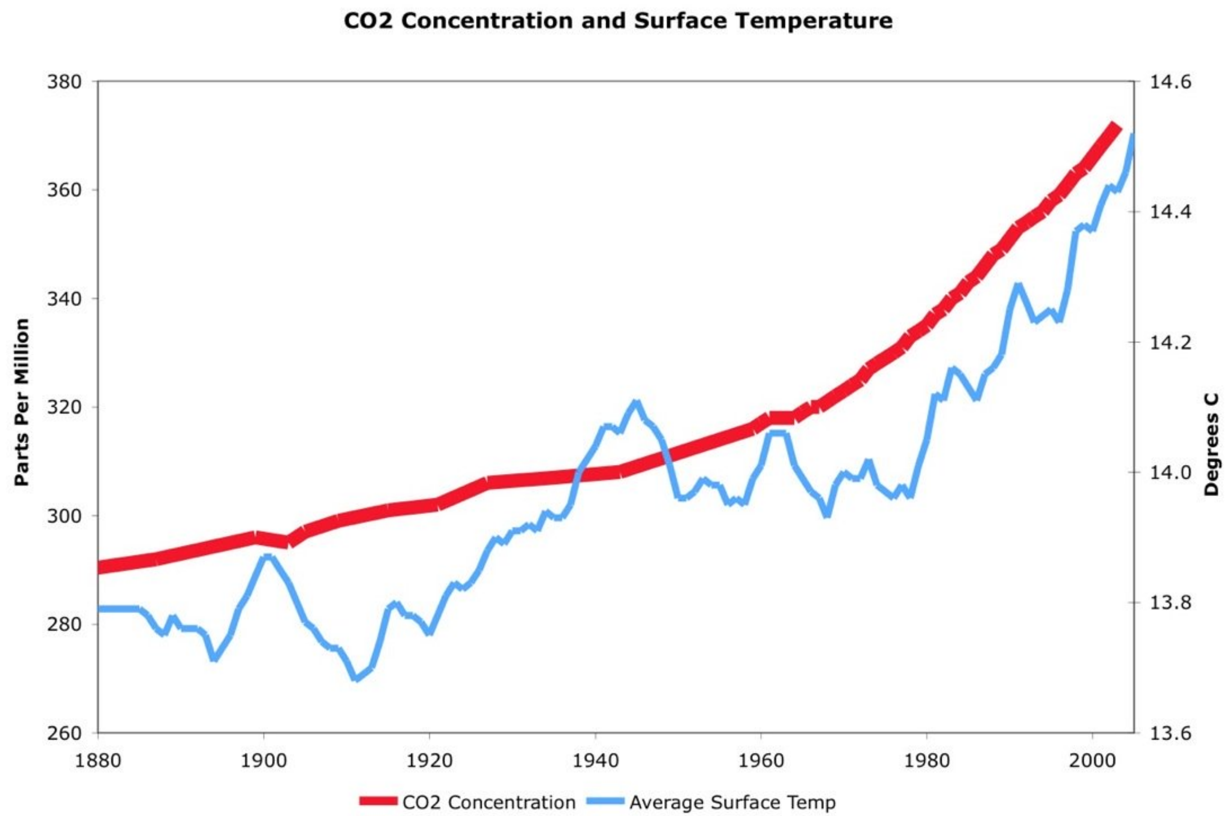
Population Growth



As Al Gore said, “Yes, they fit.” This chart shows how Total Annual Carbon Emissions are directly affected by Population Growth.

Global Environmental Concerns

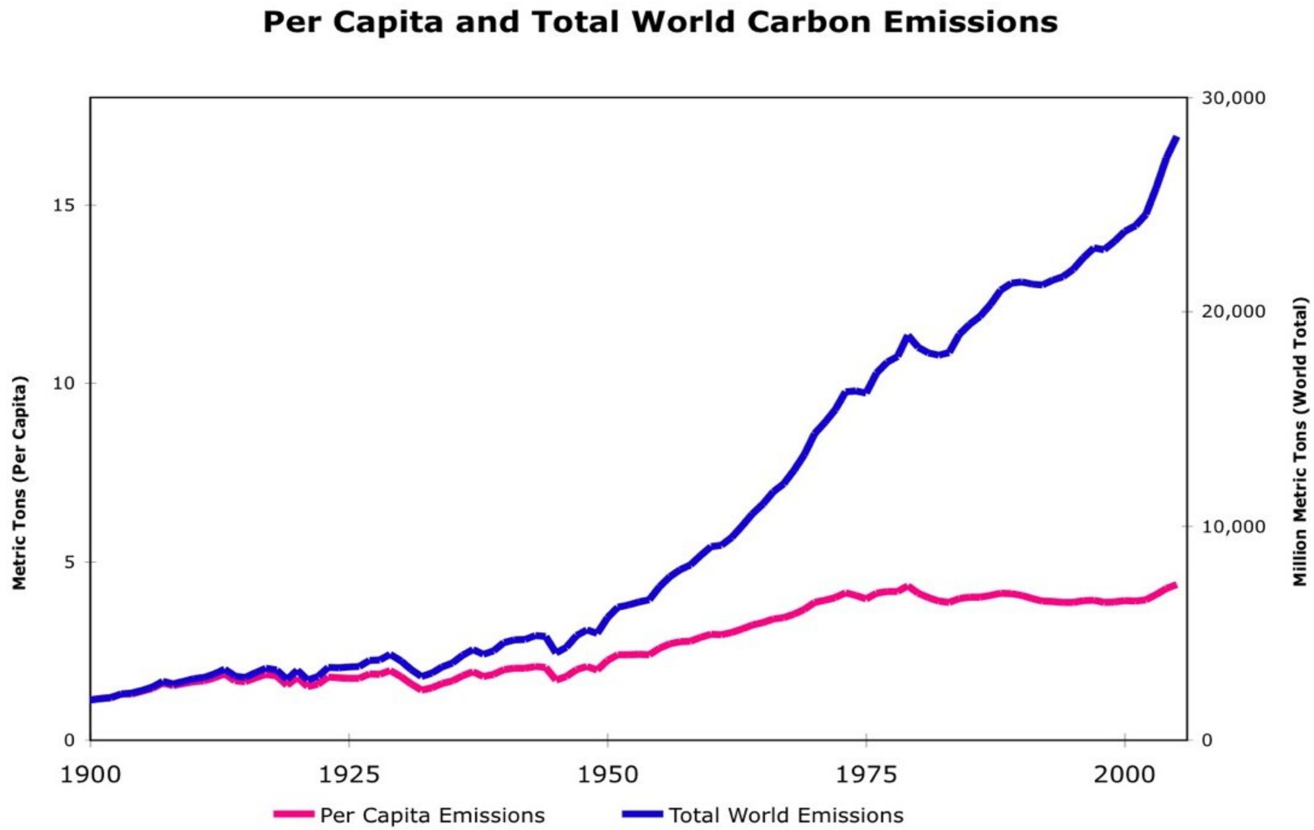
Population Growth



This chart shows the correlation between Atmospheric Carbon Concentration (in Parts Per Million) and Average Surface Temperature.

Global Environmental Concerns

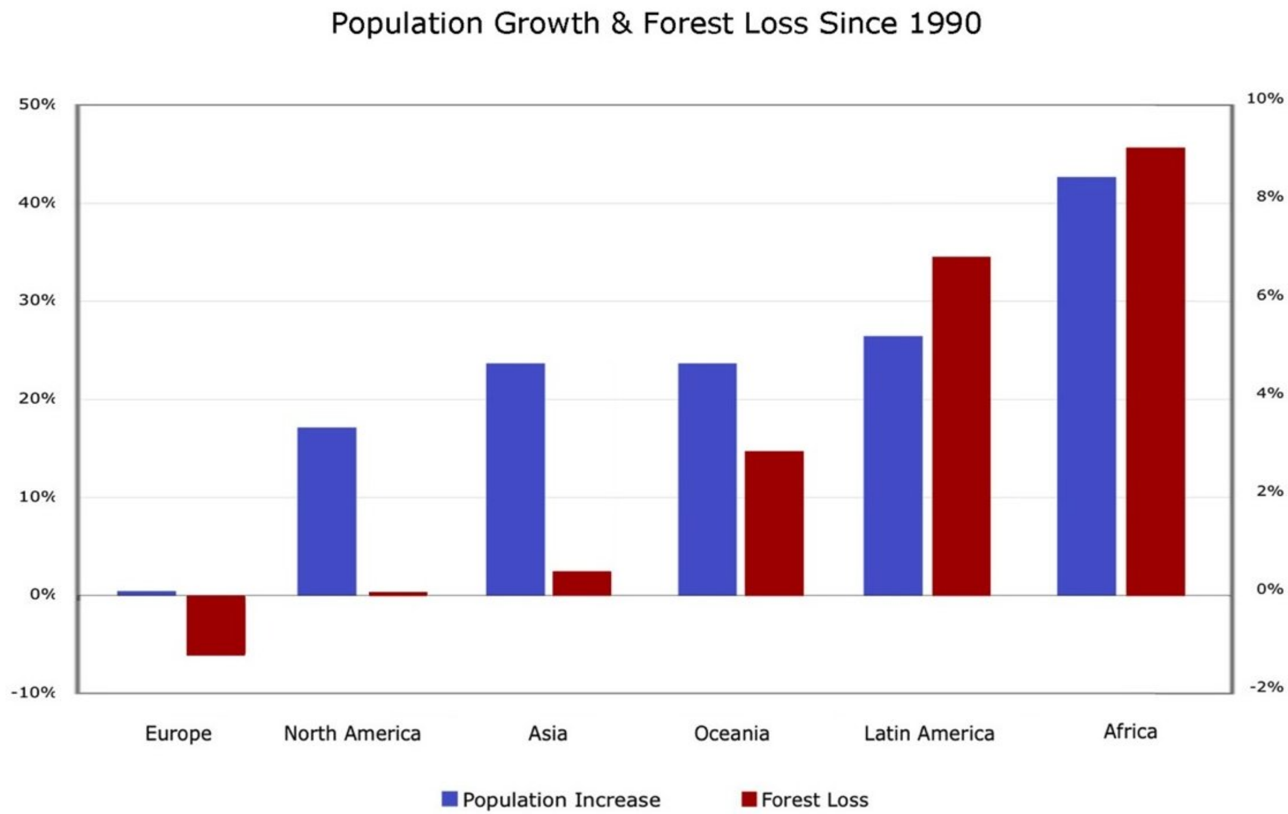
Population Growth



A curious case of divergence. This chart shows how Total Worldwide Carbon Emissions have increased at an Exponential Rate, while Per Capita Carbon Emissions have grown at a Linear Rate, during the past century. To what can we attribute the difference in rate of growth?

Global Environmental Concerns

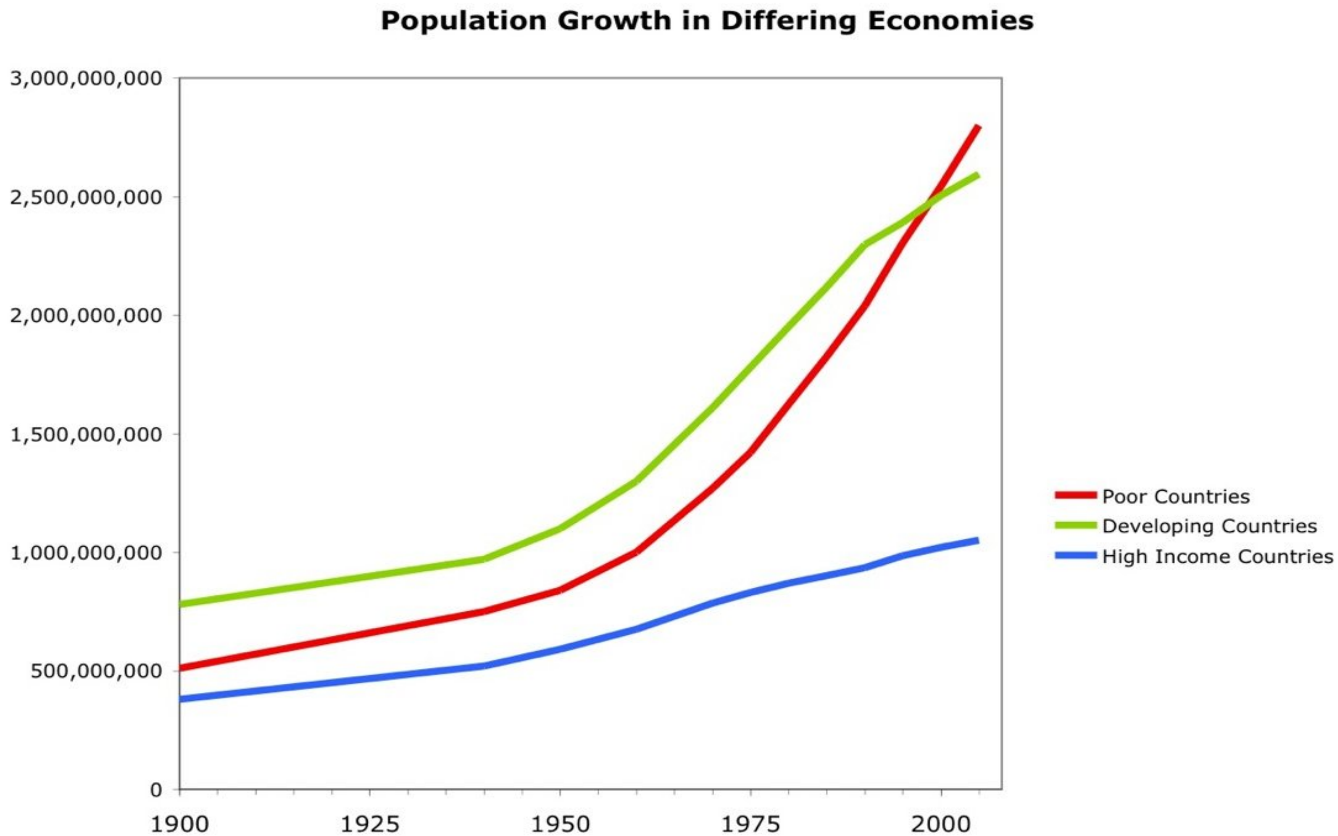
Population Growth



This very important chart shows the relationship between Population Growth and the Deforestation. Conclusion: the faster the population grows, the faster the forests disappear.

Global Environmental Concerns

Population Growth



This chart shows how Population Growth Rates are affected by Income.

Global Environmental Concerns

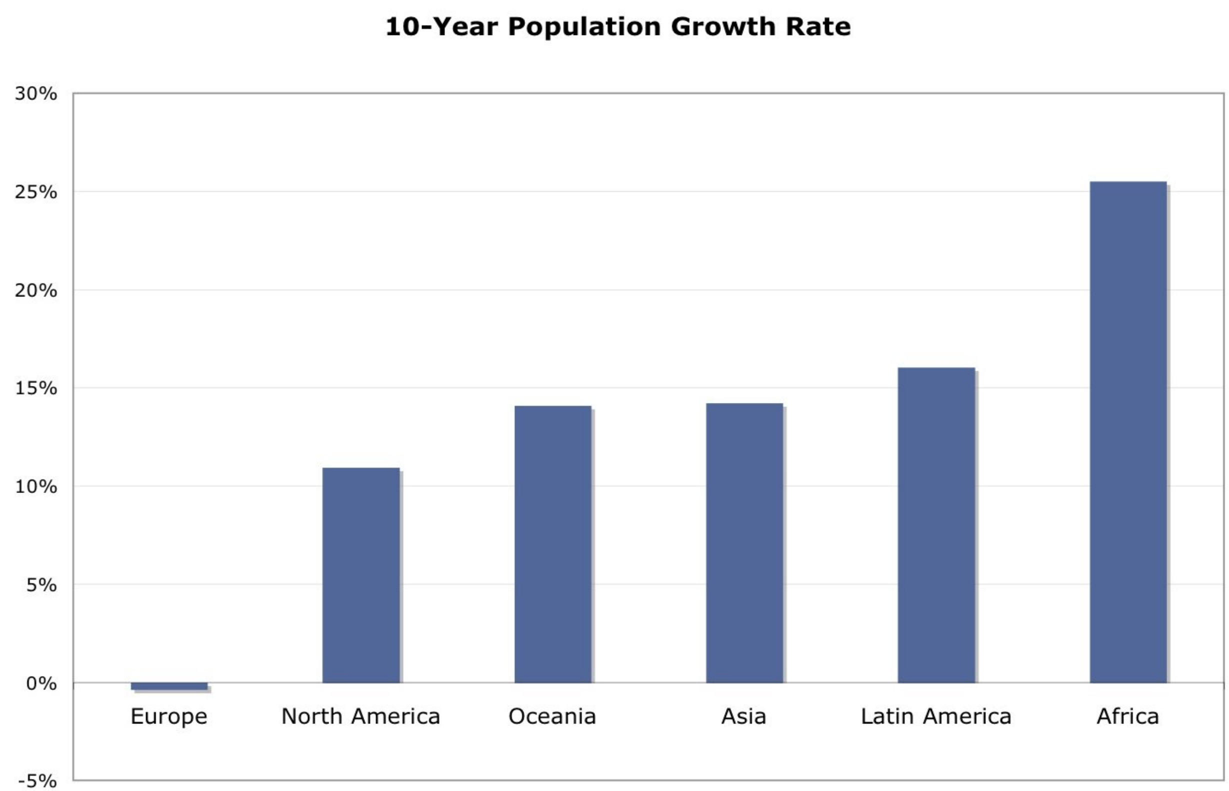
Population Growth

	2050		2010		Population Increase	
Country	Population (Millions)	Rank	Population (Millions)	Rank	Millions	Percent
India	1,748	1	1,189	2	559	47
China	1,437	2	1,338	1	99	7
United States	423	3	310	3	113	36
Pakistan	335	4	185	6	150	81
Nigeria	326	5	158	8	168	106
Indonesia	309	6	235	4	74	31
Bangladesh	222	7	164	7	58	35
Brazil	215	8	193	5	22	11
Ethiopia	174	9	85	13	89	105
Congo	166	10	68	19	98	144
Total (share of world's population)	5,355 (0.56)		3,925 (0.57)		1,430 (0.55)†	

This table shows **bulk Population** in different countries of the world.

Global Environmental Concerns

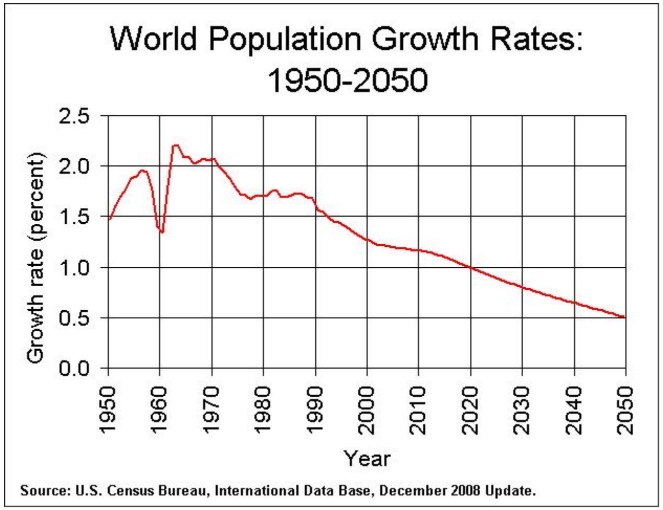
Population Growth

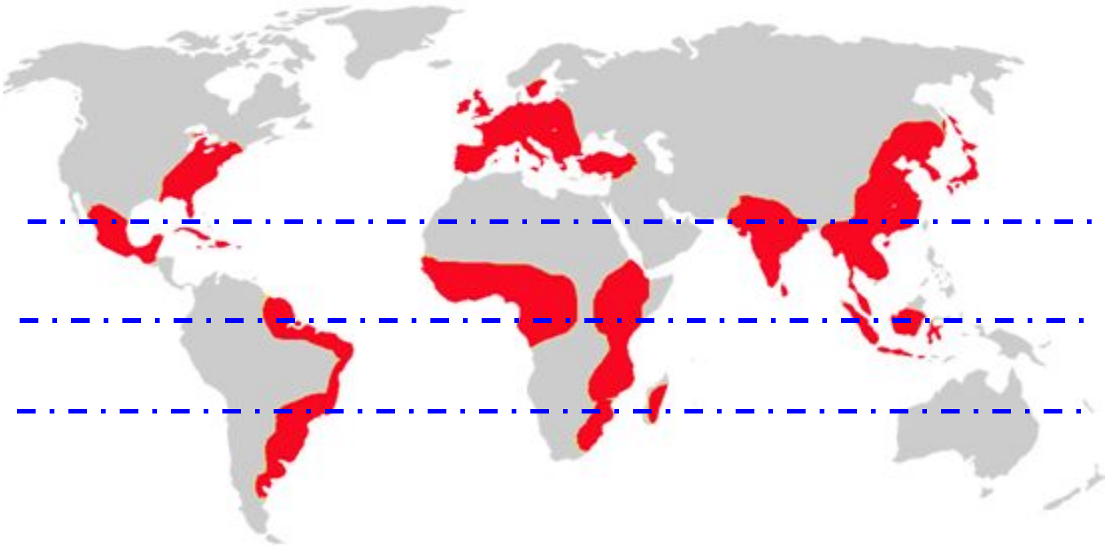


Asia has grown the most in absolute terms, **Africa** leads the world in its **Rate of Growth**.

Global Environmental Concerns

Population Growth





Global Environmental Concerns

Population Growth

Impacts of Population Growth

- great pressure is being placed on arable land, water, energy, and biological resources to provide an adequate supply of food while maintaining the integrity of our ecosystem.
- overload the earth's life-support systems
- Vital resources are stressed by the **dual demands** of increasing population and increasing consumption per person
- ground water supplies being depleted, agricultural soils being degraded, oceans being over fished, oil reserves being drawn down and forest being cut faster than they can re-grow
- human population inevitably outstrips their food supply and eventually collapse into starvation, crime and misery
- Sometimes it is viewed that the root cause of environmental degradation is inequitable distribution of wealth and power rather than population size

Global Environmental Concerns

Soil Degradation

… is a marvelous substance, a living resource of astonishing beauty, complexity, and fertility. It is a complex mixture of weathered mineral materials from rocks, partially decomposed organic molecules, and a host of living organisms.

ecosystem by itself

essential component of the biosphere

Global Environmental Concerns

Soil Degradation

Soil Degradation

based on both biological productivity and our expectations about what the land should be like. Often this is a subjective judgment, and it is difficult to distinguish between human-caused deterioration and natural conditions, like drought.

We generally consider the land degraded when the soil is impoverished or eroded, water runs off or is contaminated more then normal, vegetation is diminished, biomass production is decreased, or wildlife diversity diminishes.

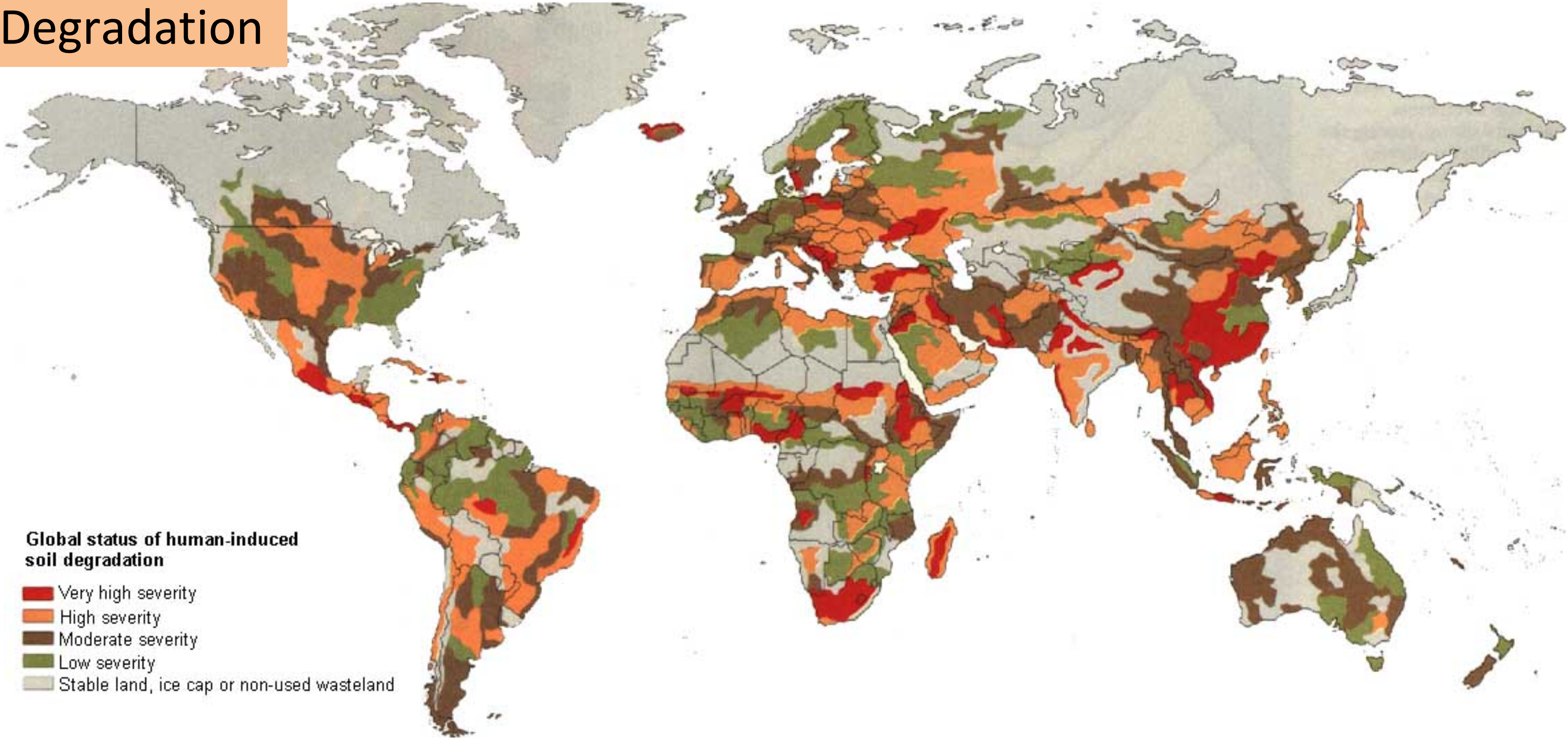
Impacts…

On farmlands, lower crop yields…. On ranchlands, fewer livestock can be supported per unit area…On nature reserves, lower biological diversity.

Global Environmental Concerns

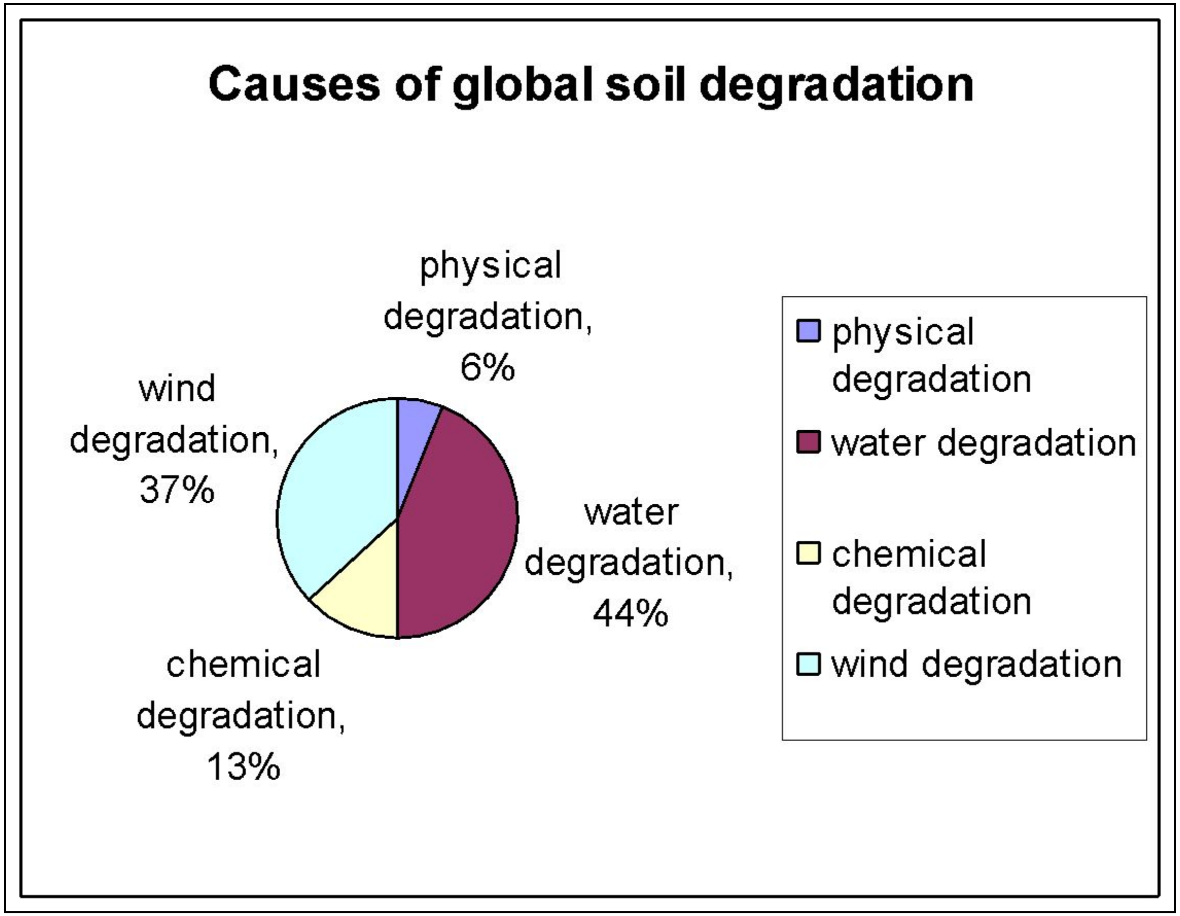
Soil Degradation

Human-induced soil degradation



Global Environmental Concerns

Soil Degradation



Global Environmental Concerns

Soil Degradation

Results…!

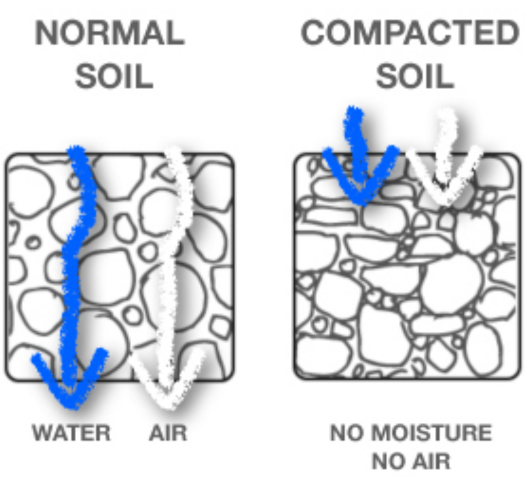
water and wind erosion

Water induced erosion in Bangladesh

Global Environmental Concerns

Soil Degradation

Physical degradation includes compaction by heavy machinery or trampling by cattle, water accumulation from excess irrigation and poor drainage, and laterization (solidification of iron and aluminum-rich tropical soil when exposed to sun and rain).



Global Environmental Concerns

Soil Degradation

chemical degradation includes nutrient depletion, salt accumulation, acidification and pollution

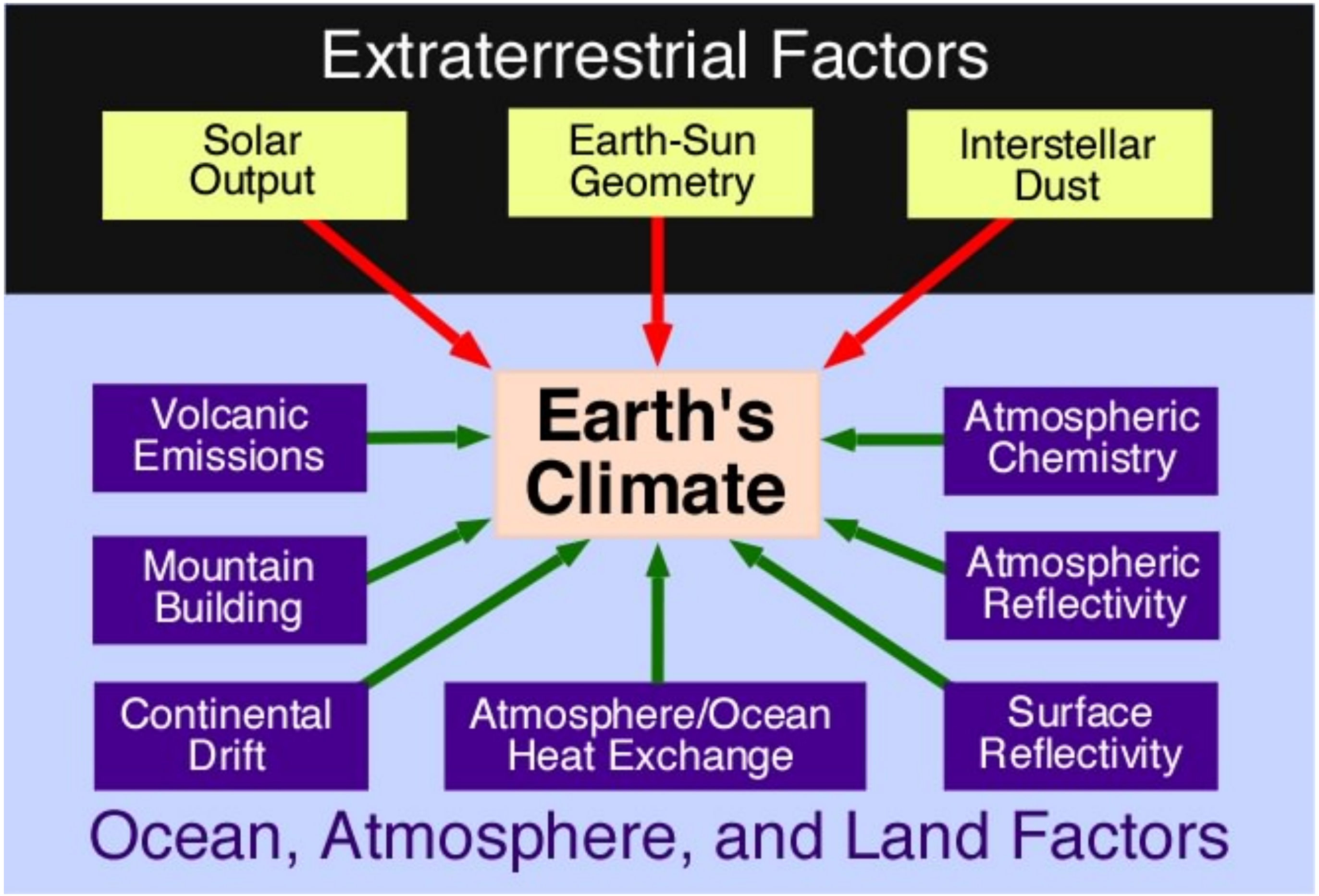
Global Environmental Concerns

Soil Degradation

sandy soil…less nutrient

Global Environmental Concerns

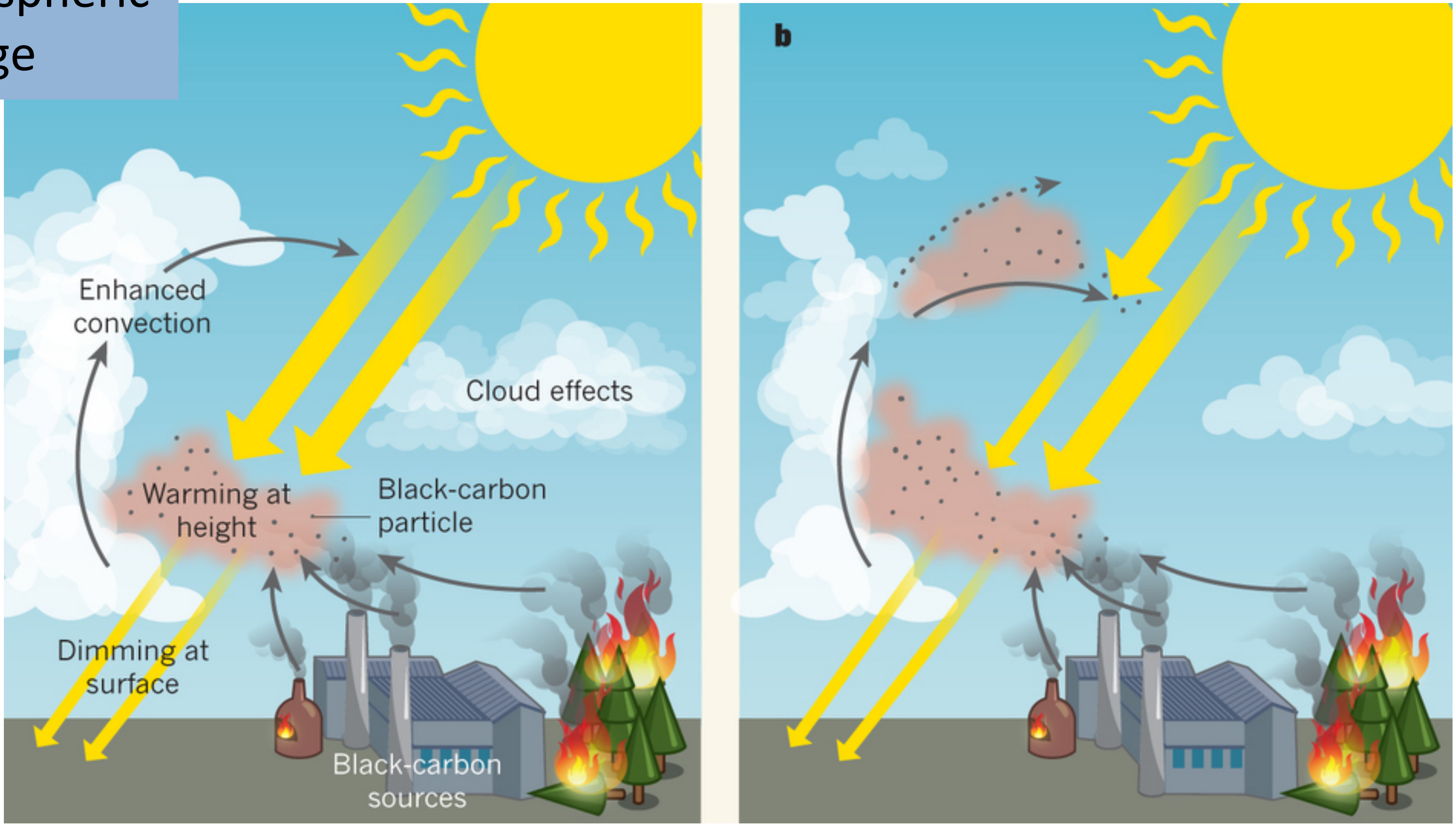
Atmospheric Change



Global Environmental Concerns

Atmospheric Change

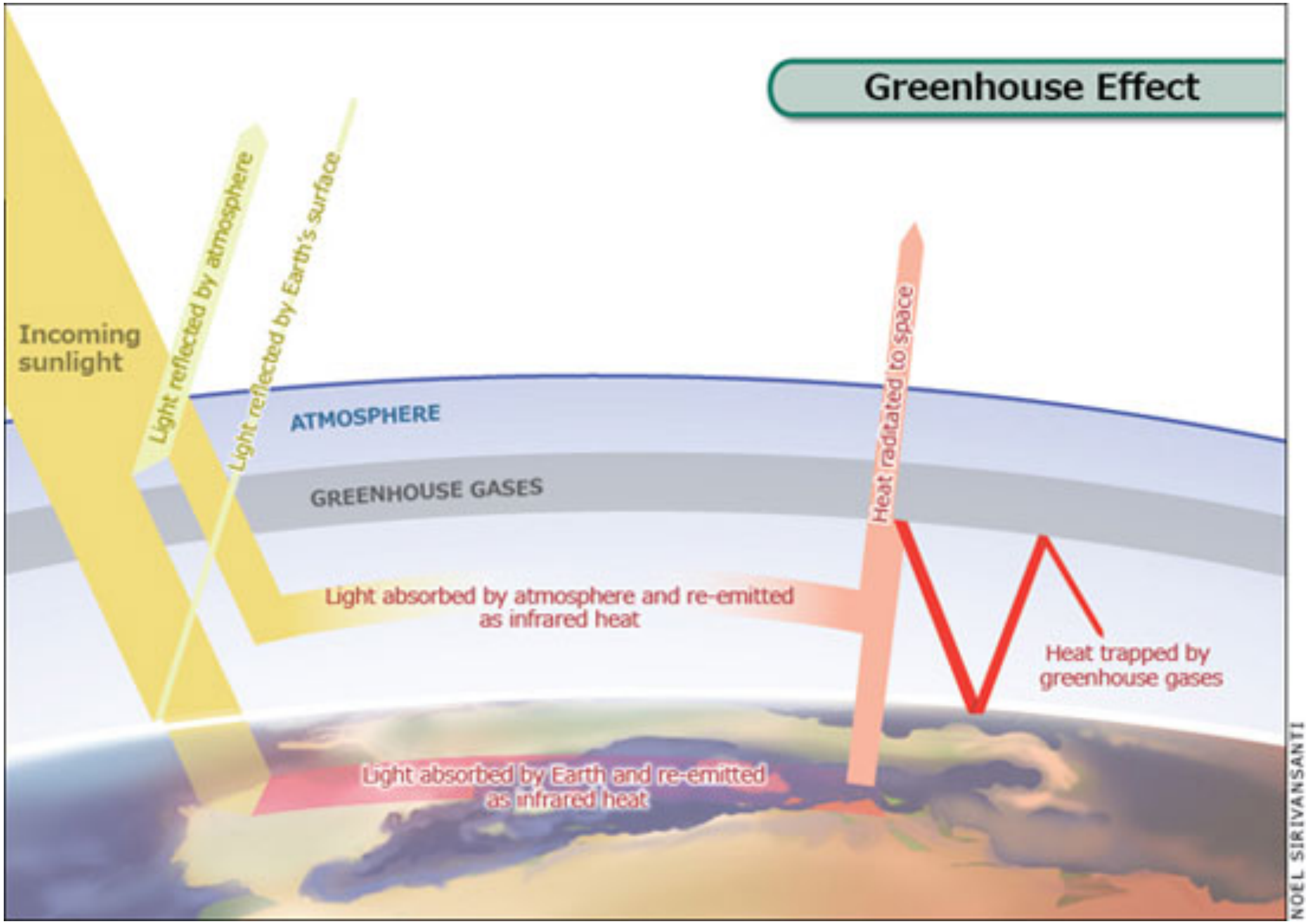
Human Induced Factors_ Anthropogenic activities



Global Environmental Concerns

Atmospheric Change

Our Concern!!!



Global Environmental Concerns

Atmospheric Change

Impacts…!!!

increased global temperature

At the current rate, global temperature may increase by 1.8–4.0 ºC by the end of this century (IPCC 2007)

Greenhouse gases can influence animal populations through mechanism … can enter terrestrial and aquatic food webs, and alter ecosystem functioning

Atmospheric contaminants commonly do this in two ways: 1. bottom-up and 2. top-down pathways

1

alter plant anatomy and physiology

2

altered population dynamics for their prey

Global Environmental Concerns

Biodiversity loss

Biodiversity loss

Defined as…

«the long-term or permanent quantitative or qualitative reduction in components of biodiversity and their potential to provide goods and services, to be measured at global, regional and national levels».

(Convention on Biological Diversity)

- Loss of diversity (e.g. species, varieties, populations, alleles etc.),
- Loss/diminished capacity for the components of diversity to provide a particular service (e.g. unsustainable harvest),
- Homogenization of biodiversity (e.g. monoculture agriculture).

Why is biodiversity loss a concern?

Life has existed on Earth for over 3.5 billion years. Over 95% of the species that ever existed have gone extinct. Why should we be concerned about current extinction rates and conserving biodiversity?

Why is biodiversity loss a concern?

Human Well-being

- Basic material for good life
- Health
- Security
- Good social relations
- Freedom of choice and action

Ecosystem Services

Provisioning Services

- Food, fiber, and fuel
- Genetic resources
- Biochemicals
- Fresh Water

Cultural Services

- Spiritual and religious values
- Knowledge system
- Education and inspiration
- Recreation and aesthetic values
- Sense of place

Supporting Services

- Primary production
- Provision of habitat
- Nutrient Cycling
- Soil formation and retention
- Production of atmospheric oxygen
- Water cycle

Regulating Services

- Invasion resistance
- Herbivory
- Pollination
- Seed dispersal
- Climate regulation
- Disease regulation
- Natural hazard protection
- Erosion regulation
- Water purification
- Pest Regulation

Ecosystem Functions

Biodiversity

- Number
- Relative abundance
- Composition
- Interactions

Millennium ecosystem assessment, 2005

Example: Provisioning services

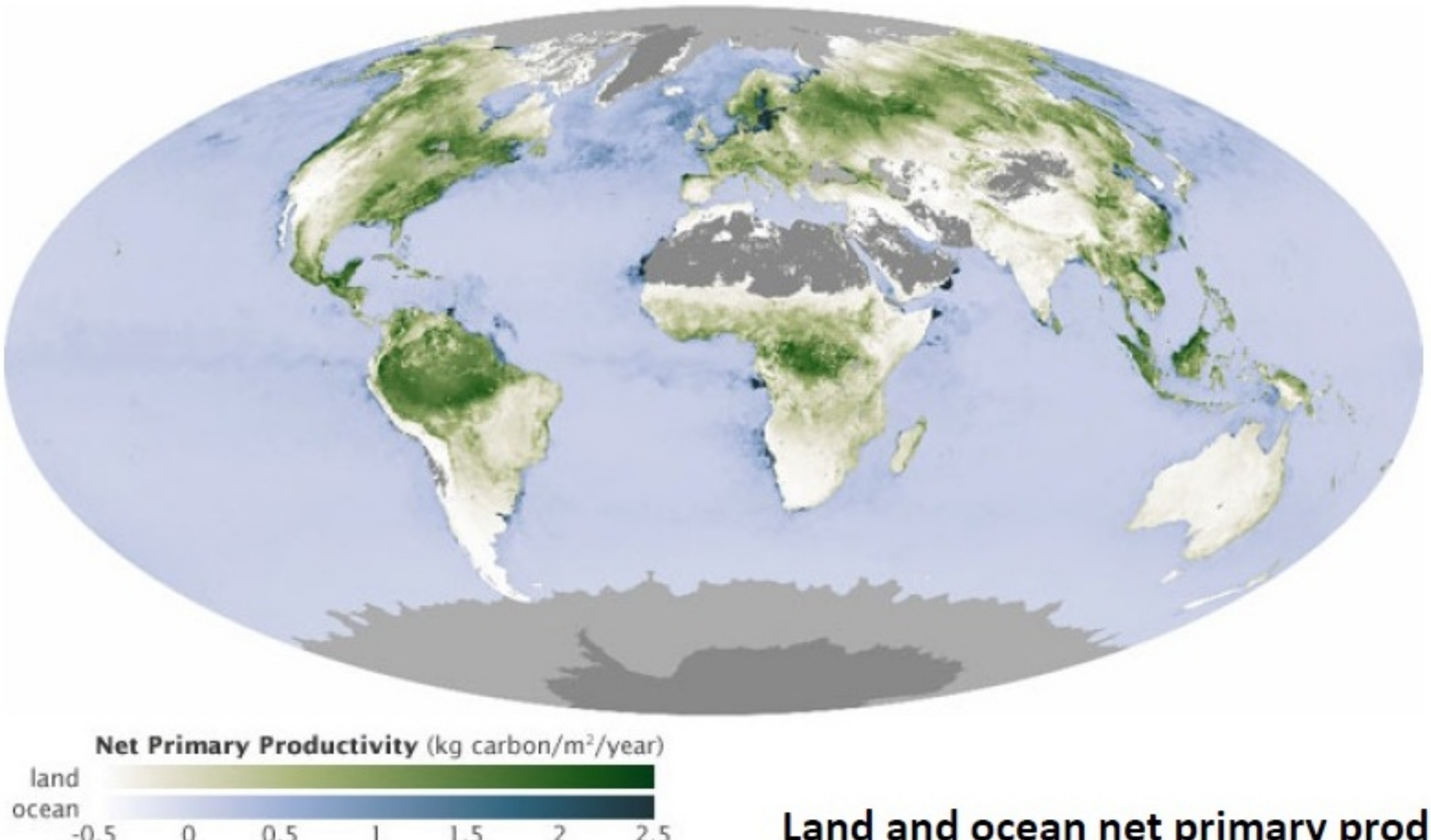
Variety of food

Example: Provisioning services

Example: Cultural services

Example: Cultural services

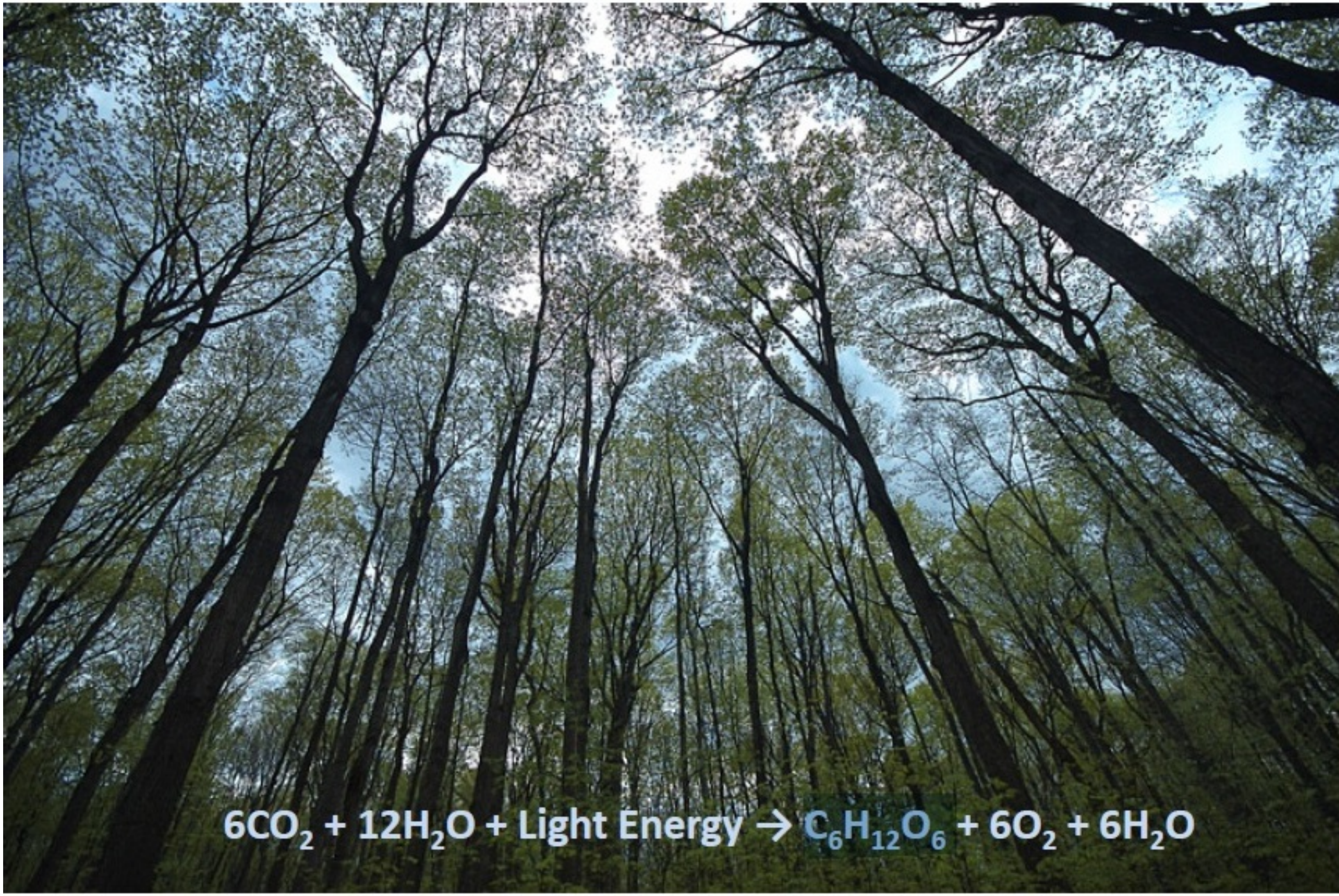
Example: Supporting service



Land and ocean net primary production estimated from spectral data

Source: NASA Earth Observatory

Example: Supporting service



Example: Regulating services

Example: Regulating service

Thanks

References

Perkins et al (2013), Economics of Development, seventh Edition

Cunningham willium, Principles of environmental, second edition

http//gadflyer.com

http://www.nature.com/scitable/knowledge/library/global-atmospheric-change-and-animal-populations